

Novelty Expression

Programmed death-ligand 1 (PD-L1) has emerged as an important biomarker in immune checkpoint inhibitor therapy for urothelial carcinoma. However, the clinicopathological significance of PD-L1 immunohistochemical expression in invasive urothelial carcinoma of the urinary bladder remains inconsistent across previous studies. Variability in antibody clones, scoring systems, and tumor heterogeneity has resulted in conflicting evidence regarding the relationship between PD-L1 expression and histopathological characteristics such as lymphovascular invasion and tumor-infiltrating lymphocytes. Furthermore, data from routine diagnostic pathology settings in developing countries remain limited.

The present study provides a clinicopathological evaluation of invasive urothelial carcinoma of the bladder with specific emphasis on the association between PD-L1 immunohistochemical expression and lymphovascular invasion, tumor-infiltrating lymphocytes, and histological subtypes defined by the **WHO Classification of Urinary and Male Genital Tumours, 5th edition (2022)**. By applying the Combined Positive Score (CPS) for PD-L1 assessment and analyzing archived formalin-fixed, paraffin-embedded specimens from routine pathological practice, this study offers additional insight into the biological significance of PD-L1 expression in invasive urothelial carcinoma.

The novelty of this study lies in demonstrating a statistically significant association between PD-L1 immunohistochemical expression and lymphovascular invasion in invasive urothelial carcinoma of the bladder, suggesting that PD-L1 expression may be more closely related to tumor invasiveness rather than immune cell infiltration. These findings contribute to the ongoing discussion regarding the clinicopathological relevance of PD-L1 expression and may provide additional evidence for understanding the role of PD-L1 as a potential biomarker in bladder cancer.